

One source of vitamin A is carotene in the diet. Carotene can be converted to vitamin A in the body.

Scientists have genetically modified a variety of rice to improve the diet of people who are vitamin A deficient. The grains of this genetically modified rice contain carotene.

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In a 2018 report, the United Nations Children's Fund (UNICEF) estimated that:

- more than 140 million children are at high risk of vitamin A deficiency
- 1.15 million child deaths are caused by vitamin A deficiency each year.

Vitamin A deficiency is also a leading cause of childhood blindness.

The recommended dietary allowance (RDA) of vitamin A for a 5-year-old child is 400 µg per day.

Genetically modified (GM) rice contains 12 µg of carotene per gram of rice. In 2018, four countries approved GM rice as safe to eat.

The development of GM rice has been partly paid for by governments. It has been agreed that seeds of GM rice will be made available to farmers in countries with high levels of poverty at the same cost as seeds of normal rice.

Some international organisations, such as Greenpeace, campaign against all GM crops.

More than 100 respected scientists have asked Greenpeace to stop campaigning against GM rice.

(b) Discuss social and ethical arguments that support the position of the scientists.

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